TASK 1 – MACHINE LEARNING CLASSIFICATION PROJECT REPORT

Customer Churn Prediction Using Machine Learning

Submitted By

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Task: Task 1 – ML Classification Project

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1. Introduction

Machine Learning classification is a supervised learning technique used to predict categorical outcomes based on historical data. Classification models are widely used in industries for customer retention, spam filtering, fraud detection, healthcare diagnosis, and recommendation systems.

The objective of this project is to build a machine learning classification model that predicts whether a customer is likely to leave a telecom service provider. Predicting customer churn helps organizations identify customers at risk of leaving and take proactive measures to improve customer retention.

In this project, the Telco Customer Churn dataset was used to train and evaluate classification models. The dataset was preprocessed by handling missing values, removing irrelevant features, and encoding categorical variables. Two machine learning algorithms, Logistic Regression and Random Forest Classifier, were implemented and compared using various evaluation metrics. The model performance was assessed using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Cross-Validation scores.

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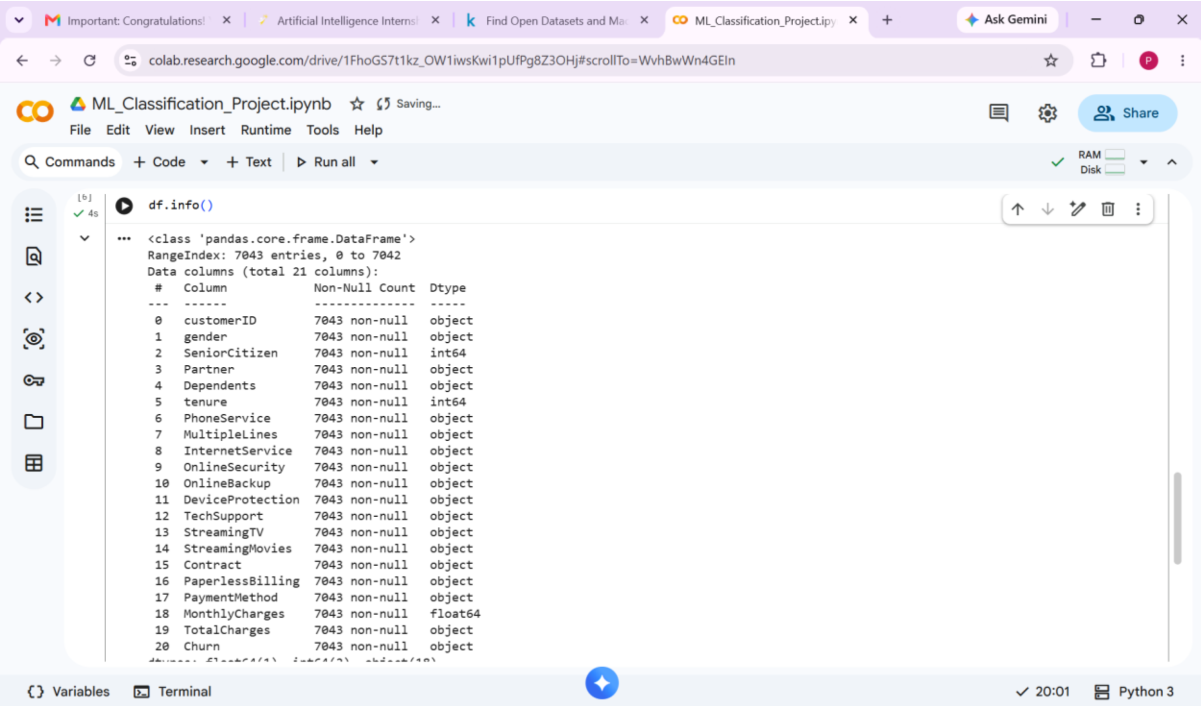
2. Dataset Information

Item	Details
Dataset Name	Telco Customer Churn Dataset
Source	Kaggle
Target Variable	Churn
Total Records	7043



```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
```

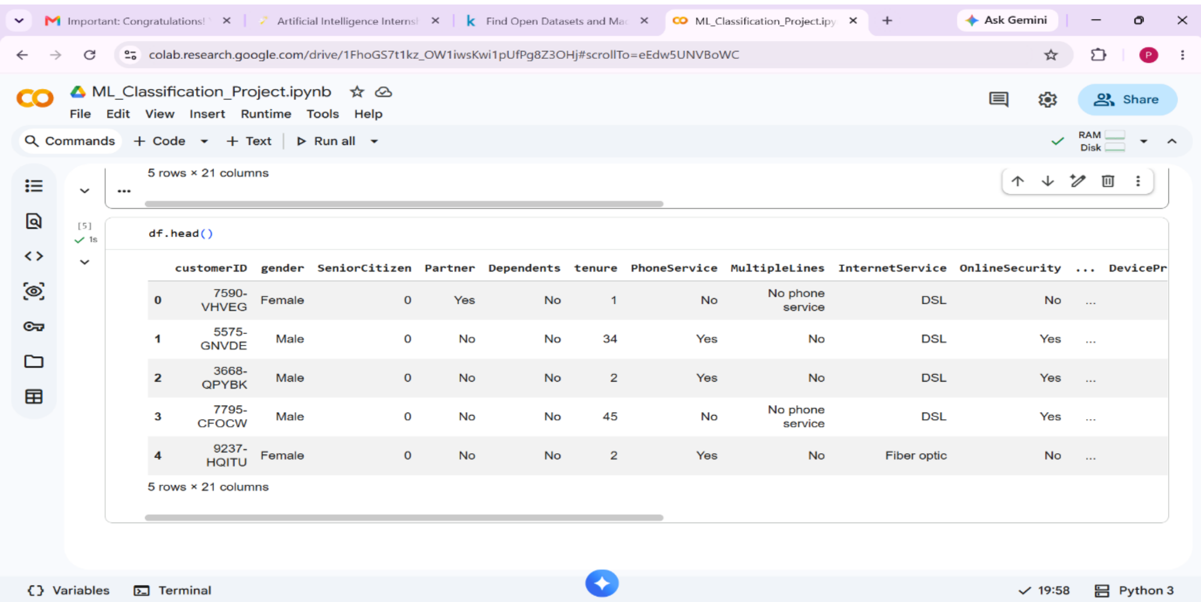
Dataset Description

The dataset contains information about telecom customers, including demographic details, account information, subscribed services, and whether the customer has churned.

Target Variable

Churn

- 1 = Customer left the company
- 0 = Customer stayed with the company



```
df.head()
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	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DevicePr
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	

5 rows × 21 columns

3. Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning algorithms.

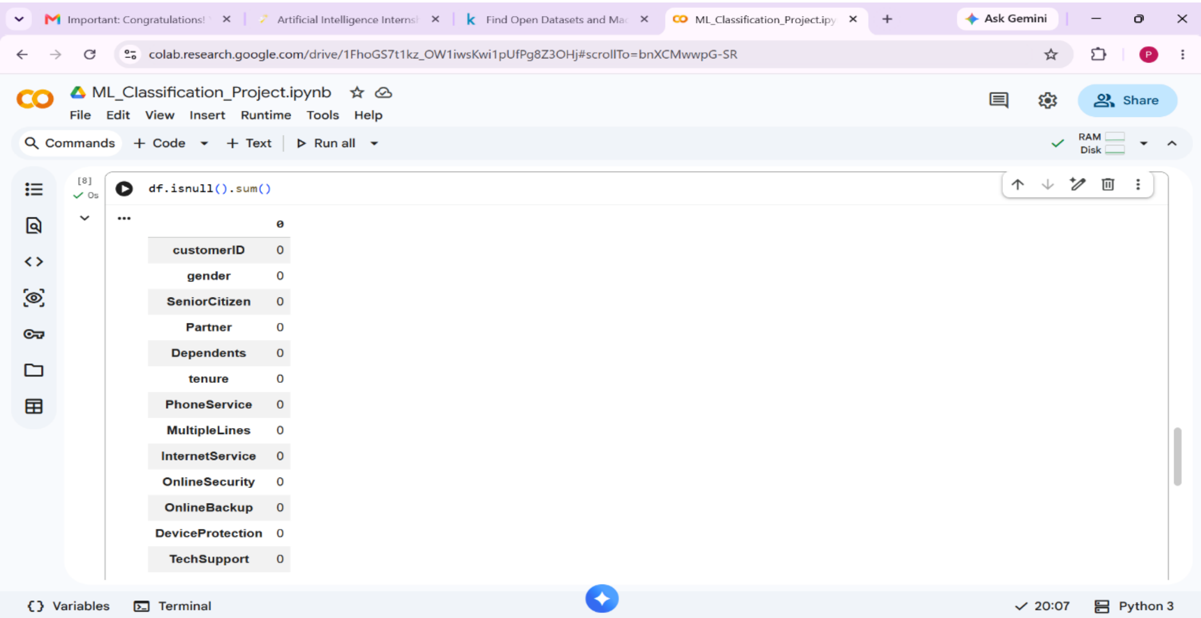
Steps Performed:

1. Handling Missing Values

The dataset was checked for missing values using:

df.isnull().sum()

Missing values in the TotalCharges column were replaced using the median value.



2. Removing Unnecessary Features

The customerID column was removed because it does not contribute to prediction.

3. Encoding Categorical Variables

Categorical features were converted into numerical values using Label Encoding.

4. Feature and Target Separation

The target variable Churn was separated from the input features.

5. Train-Test Split

The dataset was split into:

- Training Data: 80%
- Testing Data: 20%

4. Machine Learning Models Used

Two classification algorithms were implemented and compared.

4.1 Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems. It predicts the probability of a customer belonging to a particular class.

Advantages:

- Simple and interpretable
- Fast training
- Effective for binary classification

4.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

Advantages:

- Handles complex relationships
- Reduces overfitting
- Provides high predictive performance

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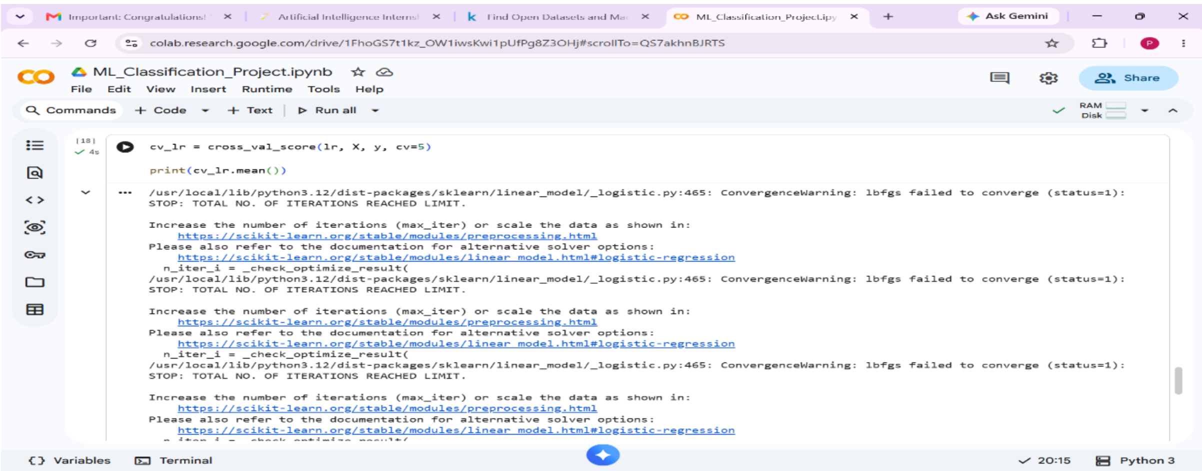
5. Cross Validation

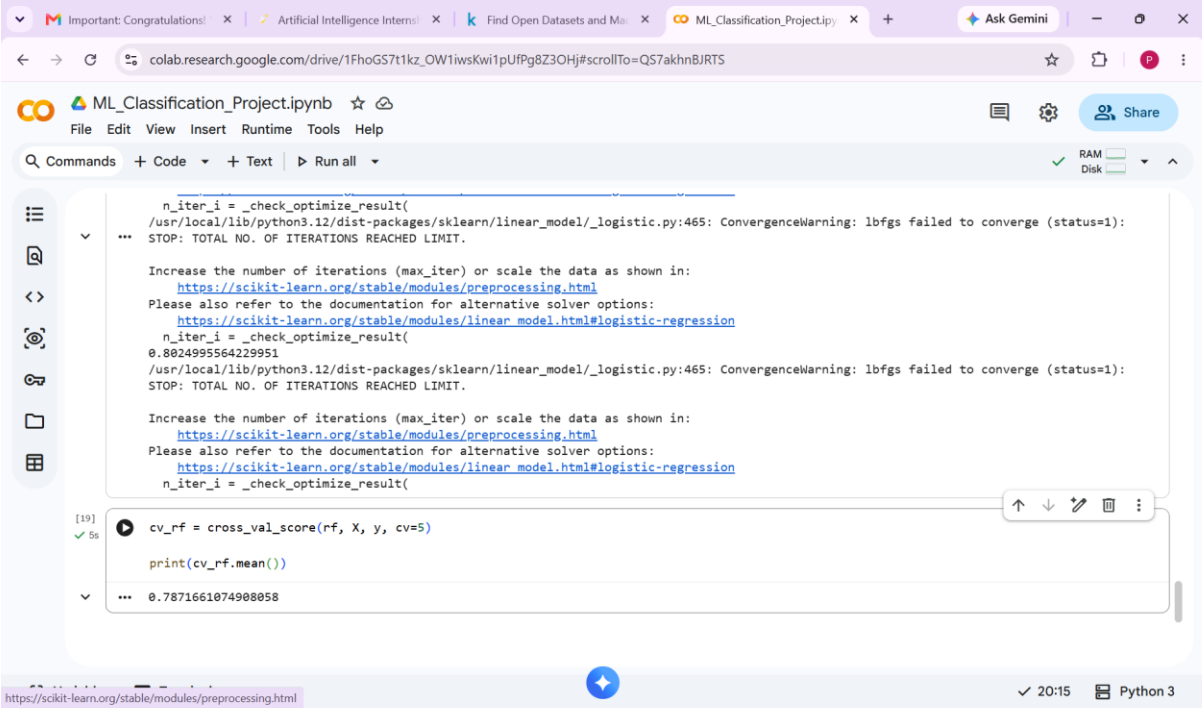
To evaluate model stability and generalization, 5-Fold Cross Validation was performed.

5.1 Logistic Regression Cross Validation Score



Mean CV Score = 80.25%

5.2 Random Forest Cross Validation Score



Mean CV Score = 78.72%

Cross-validation helps ensure that the model performs consistently across different subsets of the dataset.

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6. Model Evaluation

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

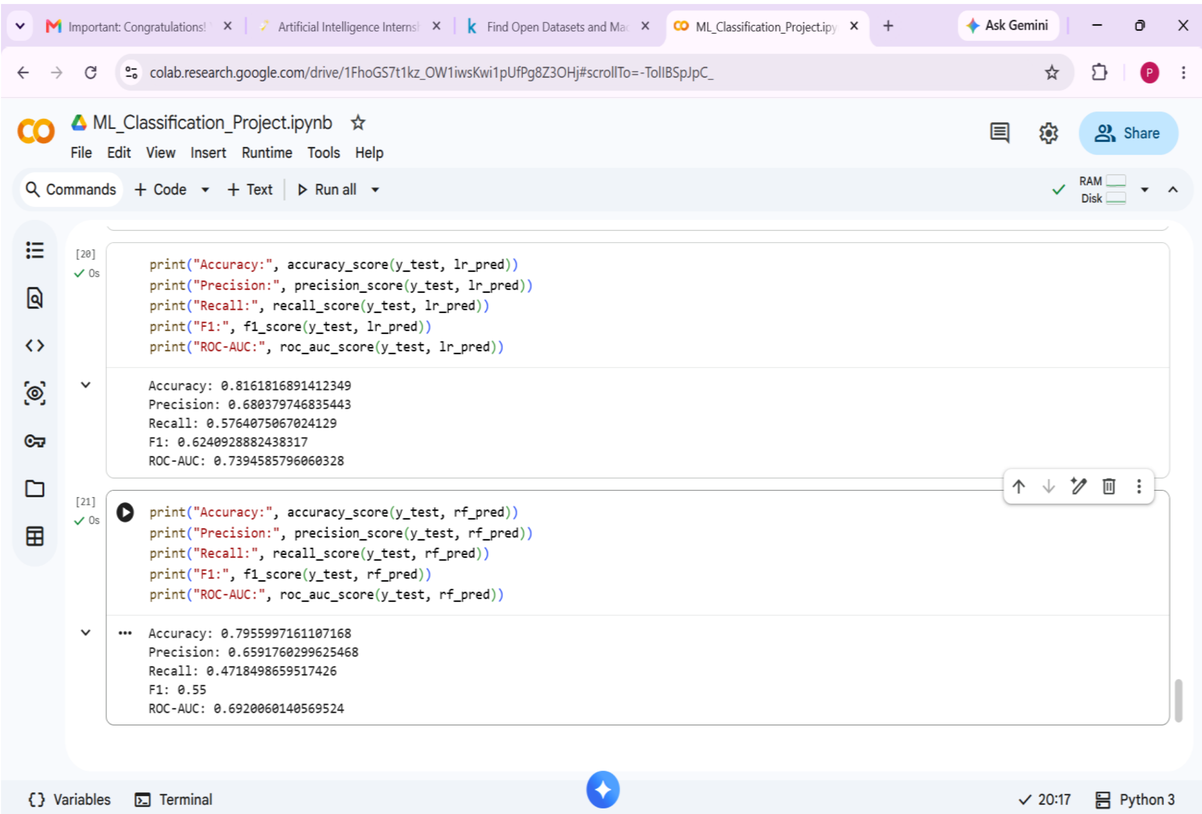
6.1 Logistic Regression Results

Metric	Value
Accuracy	0.8162 (81.62%)
Precision	0.6804 (68.04%)

Recall	0.5764 (57.64%)
F1-Score	0.6241 (62.41%)
ROC-AUC	0.7395 (73.95%)

6.2 Random Forest Results

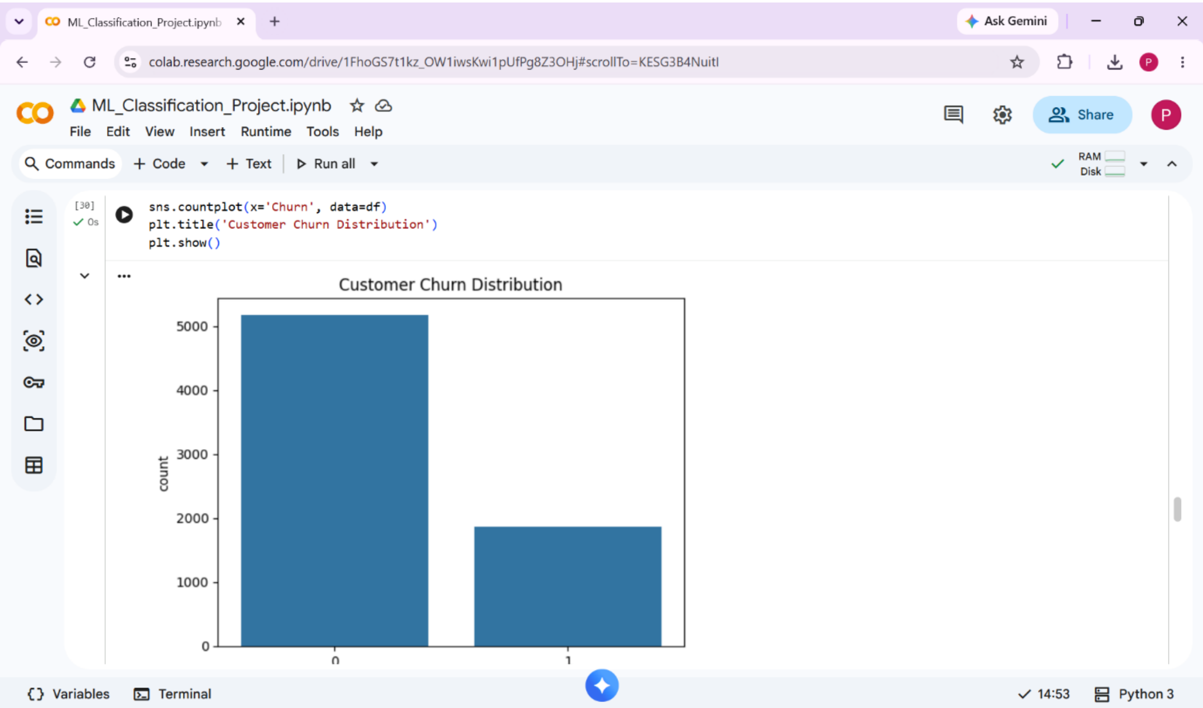
Metric	Value
Accuracy	0.7956 (79.56%)
Precision	0.6592 (65.92%)
Recall	0.4718 (47.18%)
F1-Score	0.5500 (55.00%)
ROC-AUC	0.6920 (69.20%)



7. Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	81.62%	79.56%
Precision	68.04%	65.92%
Recall	57.64%	47.18%
F1-Score	62.41%	55.00%
ROC-AUC	73.95%	69.20%
Cross Validation Score	80.25%	78.72%

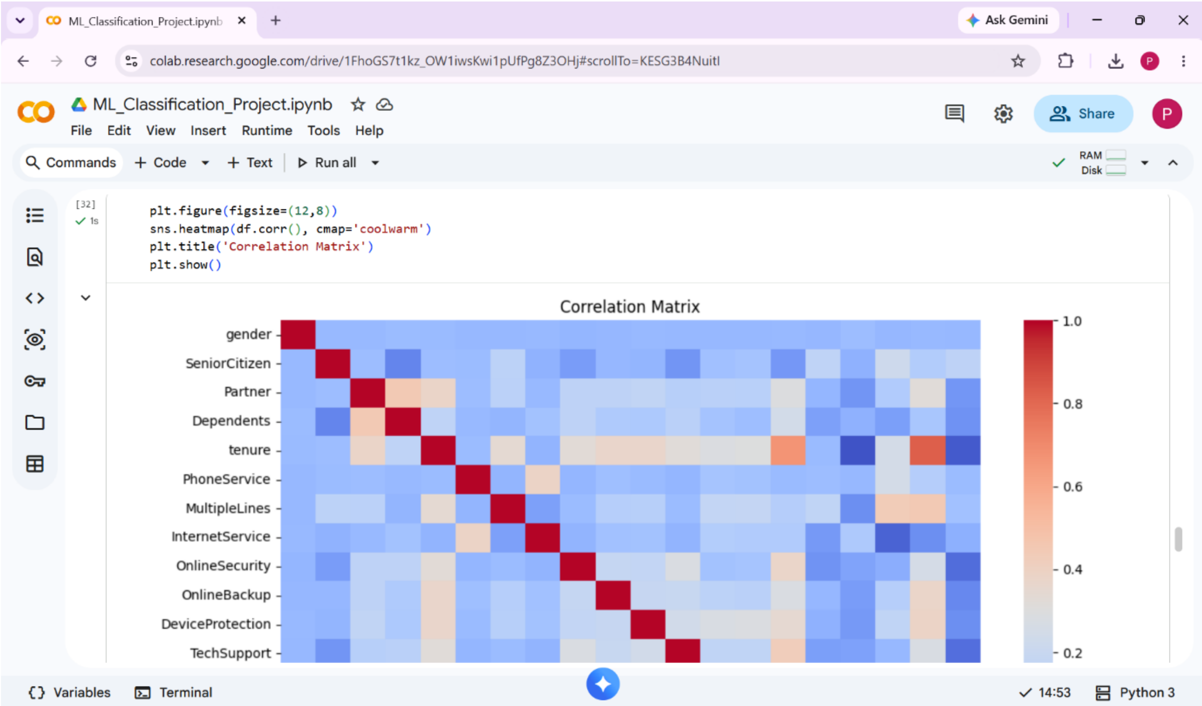
Visualizations



Analysis

The performance of both algorithms was compared using standard classification metrics. The model with higher Accuracy, F1-Score, and ROC-AUC score was considered the better-performing model.

Based on the obtained results, Logistic Regression achieved better overall performance than Random Forest Classifier because it obtained higher Accuracy, Recall, F1-Score, ROC-AUC, and Cross-Validation scores.



8. Conclusion

In this project, a machine learning classification system was developed to predict customer churn using the Telco Customer Churn dataset.

The following tasks were successfully completed:

- Data preprocessing
- Missing value handling
- Label encoding
- Train-test splitting
- Cross-validation
- Logistic Regression implementation
- Random Forest implementation
- Model evaluation and comparison

Among the evaluated models, Logistic Regression produced the best overall performance based on Accuracy, F1-Score, ROC-AUC, and Cross-Validation results.

This project demonstrates how machine learning techniques can be applied to predict customer behaviour and support business decision-making.

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9. Tools and Libraries Used

Programming Language

- Python

Development Environment

- Google Colab

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

10. References

1. Kaggle – Telco Customer Churn Dataset
2. Scikit-Learn Documentation
3. Python Documentation